

One job, three release-time decision rules

Same flexible evaluation job • same given runtime R
Same historical carbon-intensity series $I(t)$

(a) B1 runtime

CARBON INPUT

Average intensity
across the whole runtime

RESOURCE INPUT

No system-load term
in the release choice

DECISION RULE

Choose the lowest
runtime-average intensity

WAITING CONTROL

$$D \leq 4 \text{ h}$$

(b) B2 admission gate

CARBON INPUT

Intensity at candidate
release; high if $\geq q_{75}$

RESOURCE INPUT

Scheduled node requests
plus this job at release

DECISION RULE

High carbon + over cap*:
try next carbon interval

WAITING CONTROL

$$D \leq 4 \text{ h}$$

(c) Adaptive Balanced

CARBON INPUT

Runtime-average intensity
and high-carbon indicator

RESOURCE INPUT

Projected CPU/GPU class
occupancy over runtime

DECISION RULE

Minimise weighted score;
then $\geq 2\%$ lower intensity

WAITING CONTROL

$$D \leq \min(4 \text{ h}, R)$$

Release times enter Slurm; Slurm determines actual starts and allocations.

* B2 cap: $\max(1 \text{ node}, 50\% \text{ of reference high-carbon peak node requests})$.

Schematic / frozen configuration. No outcomes. Admission thresholds are not electrical power caps.